

Erect and Tape a Carton SOP

One episode erects and tapes one carton. A flat blank is squared into a tube, the four bottom flaps are folded in, three tape strips seal the bottom in an H, and the carton is turned over onto the packing spot with its open end up. The cradle is empty and the peel board is bare when the episode ends.

The order never changes: square, short flaps, long flaps, seam strip, end strip, end strip, turn over. A flap is never folded on a tube that is not square, and a strip is never laid on a seam that is not closed.

Every strip is the same three moves: the left gripper pins the carton, the right gripper lays the strip onto the seam, and the right gripper presses it down. The left arm never lays and the right arm never pins.

Nothing at the far end of the carton is reached for. Every fold, every lay, and every press happens at the carton's **right end** or along its **near side**. When the job is at the other end, the cradle is turned a half turn and the job comes to the arm. The carton comes to the work; the arm does not go to the carton.

The carton never leaves the cradle until Step 7, nothing turns further than a quarter turn while it is off the table, and no gripper opens anywhere but on the carton or on the packing spot.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the cradle in the middle, the peel board on the right, and the packing spot on the near right.
3. The whole top of the carton is in frame from above: both seating lines, all four flaps, the center seam end to end, and both end seams.
4. The near face and both end panels of the carton are visible, so every overhang can be seen being pressed down.
5. Both arms are at home with grippers open.
6. The tabletop is clear of anything but the zones listed below.
7. The right gripper reaches the **left end of the center seam** without the arm leaning out over the carton. If it does not, the cradle is moved right until it does.
8. The right gripper reaches the **far long flap's free edge** without the arm reaching to its limit. If it does not, the cradle is moved toward the front edge until it does.

9. The two arms clear each other along the seam line, with the right gripper on the seam and the left gripper on the near long flap beside it.
10. Nothing obstructs the paths between the peel board, the cradle, and the packing spot.

Materials checklist

1. One flat carton blank standing on edge in the cradle channel, leaning back against the far channel wall, its **left spine on the left seating line**.
2. The blank is a plain tube of four panels with four flaps at each end, uncrushed, with all four corner creases free enough to open to a right angle by hand.
3. The flap end standing **up** is the bottom of the carton. Both ends of the blank are alike, so no marking is needed: the end that is up is the end that gets taped.
4. The flap end resting on the channel floor is untouched all episode. Its four flaps hang in line with their panels and the tube stands on their edges.
5. Three tape strips on the peel board, sticky side down, each peeling off cleanly in one piece and each carrying a stiff folded back **grab tab** at one end.
6. One **seam strip** on the seam strip spot, longer than the carton's long side, lying left to right with its grab tab at the **right** end.
7. Two **end strips** on the end strip spots, each longer than the carton's short side, lying near to far with their grab tabs at the **near** end.
8. Each strip overhangs its seam at both ends by enough to fold down onto the panel below, and by no more than the panel is tall.
9. The cradle sits at its **home stop**, its channel is empty except for the blank, and both seating lines are visible.
10. The packing spot is bare.

Workspace layout

- **Cradle:** in the middle of the tabletop, centered between the two arms, on a turntable base. It is a channel open at both ends: a **near wall** and a **far wall** set one carton width apart, and no end walls, so a flat blank can stand in it overhanging both ends. Both walls stand no higher than half the carton, so every tape overhang can be pressed down below the strip. The carton stays in the cradle from Step 1 to Step 6.
- **Seating lines:** two lines printed across the channel floor, one near each end. A squared carton sits with an end panel on each line, centered in the channel. The lines are the definition of "centered" and they read the same after a half turn.
- **Turn tabs:** two tabs on the turntable base, one at its **near right** corner and one at its **far left** corner. A half turn brings the far left tab to the near right corner, so a tab always sits where the right gripper can take it. The base has two hard **turn stops**, half a turn apart.
- **Peel board:** on the right, within reach of the right arm, holding the three strips in the order they are used, nearest the cradle first: the seam strip, then the two end strips in a row

outboard of it. Its surface releases a strip cleanly when the grab tab is lifted.

- **Packing spot:** on the near right of the tabletop, bare. The finished carton is set here and nothing else is ever set on it.
- **Left side of the tabletop:** kept clear. The left arm's whole working area is the carton itself, approached from the left. Nothing is ever staged or set down there.

Vocabulary

- **Arm assignment:** the left arm holds. It braces the blank while it is squared and pins the carton for every fold and every strip. The right arm makes every push, every fold, every turn, every lay, and every press. Neither arm takes the other's job. The one shared motion in the episode is the turn over in Step 7, where both grippers hold the carton at once and neither works alone.
- **Reach rule:** the left arm's contacts are the carton's **left end** and its **middle**. The right arm's contacts are the carton's **right end** and its **near side**. The only motion that runs the whole width of the carton is the seam press stroke in Step 4, which travels left to right at one constant depth and never leans out over the carton.
- **Blank:** the flat carton, folded shut, before it is squared.
- **Spine:** one of the two folded edges of the flat blank, at its left end and its right end. Each spine becomes a corner of the squared carton.
- **Tube:** the blank once it is square: four panels, four corners at right angles, open at both ends.
- **Squared:** near and far panels flat against the channel walls with no bow, all four corners open to a right angle, and an end panel sitting on each seating line.
- **Bottom end:** the flap end facing up. It is the only end that is folded and taped.
- **Top end:** the flap end resting on the channel floor. Nothing is done to it, and after Step 7 it is the end that faces up for packing.
- **Long flaps:** the two wide bottom flaps, on the carton's **near** and **far** sides. They meet at the center seam.
- **Short flaps:** the two narrow bottom flaps, at the carton's **left** and **right** ends. They fold in first and end up trapped under the long flaps.
- **Center seam:** the line where the two long flaps meet, running left to right along the middle of the carton's bottom.
- **End seam:** the line where the folded long flaps meet an end panel, running near to far at each end of the carton. There are two.
- **Fold:** the right gripper pinching a flap at the middle of its free edge, swinging it in until it lies flat and level, holding it there for 2 seconds, and releasing.
- **Brace:** the closed left gripper pressed against a spine and holding it still while the right gripper pushes the other spine toward it. A brace never travels.

- **Pin:** the closed left gripper pressed straight down on the carton to hold it still. A pin never pinches and never travels, and it is held for the whole step.
- **Half turn:** the right gripper pinching the **near right turn tab** and swinging it along the near edge to the **left**, in one continuous motion, until the base meets its stop. Every half turn goes the same way, every time. Both grippers are clear of the carton while it turns.
- **Strip:** one length of tape, cut to size. It is only ever handled by its **grab tab**.
- **Grab tab:** the folded back end of a strip, with no tack on it. It is the only part of a strip a gripper ever touches.
- **Overhang:** the part of a strip that hangs past the end of its seam and is pressed down onto the panel below it.
- **Hangs straight:** the strip dangling free and flat from the lifted grab tab, not curled, twisted, or stuck to itself. A strip that is not hanging straight is scrapped, not straightened.
- **Lay:** the right gripper lowering the strip's free end onto the start of a seam, then travelling along the seam and letting the strip settle onto it behind the gripper, and releasing the tab at the far end.
- **Press stroke:** the closed right gripper pressed down on a laid strip and drawn along it in one continuous stroke, end to end, without lifting.
- **Quarter turn:** both grippers pressing on the carton's two end panels, lifting it straight up clear of whatever it is standing on, rolling it toward the front edge through one quarter turn about the line between the two grippers, and lowering it until it rests. The carton is turned over by two quarter turns with a set down between them, never by one continuous half turn.
- **Settled:** the carton or flap stays put for 2 seconds after the gripper lifts clear, with nothing springing back, rocking, or sliding.
- **Release point:** on the carton, and the packing spot. A gripper opens nowhere else, and never over the peel board or the bare tabletop.

Steps

Steps 1 to 7 are one carton. Every fold is taken at the middle of a flap's free edge and held for 2 seconds before the gripper opens, and every strip is carried in the orientation it will be laid in.

Step 1: Square the blank into a tube

Goal: the blank standing in the channel as a square tube, an end panel on each seating line.

- The **left gripper**, closed, braces the **left spine** on the **left seating line**, pressing it from the left, and holds it there for the whole step.
- The **right gripper**, closed, presses on the **right spine** and pushes it straight along the channel toward the left spine, in one continuous motion, until the near and far panels lie flat against the channel walls and the **right end panel sits on the right seating line**.
- Both grippers lift clear.

Push along the channel, not across it. A push angled into the far wall folds the blank flatter instead of opening it, and the corner creases set the wrong way.

Check: all four corners are open to right angles, the near and far panels are flat against the channel walls with no bow, both end panels sit on their seating lines, all four bottom flaps stand up in line with their panels, and the tube holds its shape for 2 seconds after both grippers leave. If a corner is still creased flat, a panel bows, or the tube springs back, brace the left spine again and push the right spine the rest of the way.

Step 2: Fold the two short flaps in

Goal: both short flaps folded flat and level across the bottom end, one at each end of the carton.

- The **left gripper**, closed, pins the **near panel's outer face at the middle of the carton**, holding it against the far channel wall, and holds it there.
- The **right gripper** pinches the **right short flap** at the middle of its free edge, swings it in over the open end until it lies flat and level, holds it for 2 seconds, and releases.
- Both grippers lift clear of the carton. The **right gripper** pinches the **near right turn tab** and swings it left in one motion until the base meets its stop. Release the tab and lift clear.
- The **left gripper** pins the near panel at the middle again. The **right gripper** folds the short flap now at the right end the same way.
- Both grippers lift clear.

The cradle turns; the arm does not travel. Reaching across the carton for the far flap is what pulls the tube out of square and what makes the arm stretch.

Check: both short flaps lie flat and level across the bottom end, neither standing up, curling, nor overlapping the other, and the tube is still square with both end panels on their seating lines. If a flap has sprung back up, fold it again and hold it longer before releasing.

Step 3: Fold the two long flaps down

Goal: both long flaps down and meeting edge to edge along the center seam, with the short flaps trapped flat underneath.

- The **left gripper**, closed, pins the **left short flap** flat and holds it there for the whole step.
- The **right gripper** pinches the **far long flap** at the middle of its free edge, swings it toward the front edge until it lies flat and level over both short flaps, holds it for 2 seconds, and releases.
- The **right gripper** pinches the **near long flap** at the middle of its free edge, swings it away from the front edge until its free edge meets the far flap's free edge along the middle of the carton, holds it for 2 seconds, and releases.
- Both grippers lift clear.

The far flap goes first. With the near flap already down, the gripper has to travel out over it to reach the far one, which drags the near flap out of line and puts the arm exactly where it should never be.

Check: both long flaps lie flat and level, their free edges meet along the middle of the carton in one straight seam with no overlap and no gap the seam strip cannot cover, both short flaps are trapped flat underneath with no corner sticking out, and nothing is standing above the flap line. If the flaps overlap or a corner stands up, fold the near flap back up and lay it down again.

Step 4: Lay and press the seam strip

Goal: the seam strip down the whole center seam, pressed flat, with its right overhang down.

- The **left gripper**, closed, pins the **near long flap at its left end**, clear of the seam, and holds it there for the whole step.
- The **right gripper** pinches the **grab tab** of the **seam strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it to the carton without turning it.
- Lower the strip's free end onto the seam at the carton's **left end**, so the overhang stands past the left end panel. Travel right along the seam, letting the strip settle onto the seam behind the gripper, until the tab end lies past the right end panel. Release the tab.
- The **right gripper**, closed, presses down on the strip at the left overhang and draws along the seam in one continuous stroke, left to right, to the right overhang, without lifting.
- The **right gripper** presses the **right overhang** down onto the right end panel.
- Both grippers lift clear.

The left overhang is not pressed here. It is pressed in Step 6, after the half turn brings it to the right. Nothing is reached for.

Check: the strip runs down the middle of the seam end to end, covering it along its whole length, lying flat with no bubble, no fold, and no wrinkle, and the right overhang is flat against the end panel. If the strip landed off the seam, folded on itself, or lifted at an edge, peel it off, scrap it, and lay a fresh seam strip. A strip is never peeled up and stuck down again.

Step 5: Lay and press the first end strip

Goal: an end strip across the right end seam, pressed flat, with both overhangs down.

- The **left gripper**, closed, pins the **center seam at the middle of the carton** and holds it there for the whole step.
- The **right gripper** pinches the **grab tab** of the first **end strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it to the carton without turning it.
- Lower the strip's free end onto the **right end seam** at the far side, so the overhang stands past the far panel. Draw toward the front edge, letting the strip settle onto the seam behind the gripper, until the tab end lies past the near panel. Release the tab.

- The **right gripper**, closed, presses down on the strip at the far overhang and draws along the end seam in one continuous stroke, toward the front edge, to the near overhang, without lifting.
- The **right gripper** presses the **far overhang** down onto the far panel, then the **near overhang** down onto the near panel.
- Both grippers lift clear.

Check: the strip covers the right end seam end to end, crossing the seam strip in the middle, lying flat with no bubble, fold, or wrinkle, and both overhangs are flat against their panels. If the strip landed off the seam or folded on itself, peel it off, scrap it, and lay the second end strip in its place, then take a fresh strip for Step 6.

Step 6: Turn the cradle and finish the second end

Goal: the H complete: the second end strip down, and every overhang pressed.

- Both grippers are clear of the carton. The **right gripper** pinches the **near right turn tab** and swings it left in one motion until the base meets its stop. Release the tab and lift clear.
- The **right gripper** presses the seam strip's remaining overhang, now at the right end, down onto its end panel.
- The **left gripper**, closed, pins the **center seam at the middle of the carton** and holds it there.
- The **right gripper** lays and presses the second **end strip** across the end seam now at the right, exactly as in Step 5: hang it straight, lower the free end at the far side, draw toward the front edge, release the tab, press far to near in one stroke, then press the far overhang and the near overhang down.
- Both grippers lift clear.

The second end is the same motion as the first, not a mirror image of it. That is what the turn buys.

Check: the bottom shows one strip down the center seam and one strip across each end seam, forming an H; all four seam strip and end strip overhangs are flat against their panels; no seam line is visible through a gap between strips; and the carton is still square with both end panels on their seating lines. If an overhang has lifted, press it down again with the closed gripper.

Step 7: Turn the carton over onto the packing spot

Goal: the carton standing on its taped bottom at the packing spot, open end up.

- The **left gripper**, closed, presses on the **left end panel** and the **right gripper**, closed, presses on the **right end panel**, both at the middle of the panel's height.
- Both grippers lift the carton straight up until it is clear of the channel walls.

- Both grippers roll the carton **toward the front edge** through one **quarter turn**, keeping hold throughout, until the taped bottom faces the front edge, and lower it onto the packing spot so it comes to rest lying on its near panel. Release both grippers and lift both clear.
- Both grippers press on the **left end panel** and the **right end panel** again and lift the carton straight up clear of the table.
- Both grippers roll it **toward the front edge** through a second **quarter turn**, until the taped bottom faces down, and lower it straight down onto the packing spot until it is resting. Release both grippers and lift both clear.

Set it down between the two turns. One continuous half turn runs both wrists out to their limit, and an arm that stalls part way round drops the carton or drags it across the spot.

Check: the carton stands square on its taped bottom at the packing spot, open end up, all four top flaps standing clear of the opening, no strip lifted or torn by the handling, and settled. If it lands on an edge or off the spot, take both end panels again, lift it clear, and lower it again.

Step 8: End the episode

- Confirm the carton stands taped side down on the packing spot, the cradle channel is empty and back at its home stop, and the peel board is bare.
- Return both arms home, then stop recording.

Two half turns per episode bring the cradle back to the stop it started from. If it is not there, a turn was missed or run past its stop.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the taped carton off the packing spot and set it on the finished stack, off the tabletop. A taped carton is not reused.
2. Confirm the cradle sits at its home stop. If it does not, turn it to the stop by hand.
3. Clear the channel floor and the tabletop of tape scraps, backing, and any scrapped strip.
4. Stand a fresh flat blank on edge in the channel, leaning back against the far wall, left spine on the left seating line, with the end that will be taped facing up.
5. Reload the peel board: one seam strip on the seam strip spot lying left to right with its grab tab at the right, and two end strips on the end strip spots lying near to far with their grab tabs toward the front edge.
6. Check every grab tab is still folded back and free of tack, and that each strip lifts off the board in one piece.
7. Check the cradle: both channel walls firm, both seating lines visible, both turn tabs attached, and both turn stops holding the base square.

8. Replace any item that is damaged: a blank that is crushed, torn, or too stiff to square, a blank whose long flaps will not meet at the seam, a strip that has lost its tack, or a peel board that no longer releases a strip cleanly.
9. Wipe the tabletop and confirm the cradle and the peel board sit firm and have not shifted.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a flap is folded before the tube is square, a long flap goes down before both short flaps are in, the near long flap goes down before the far one, a strip is laid before the flaps are down, or an end strip is laid before the seam strip.
- **SOP rule broken:** Steps 1 to 6, the order never changes and a strip is never laid on a seam that is not closed.
- **Coaching note:** square, short, long, seam, end, end. Nothing jumps the queue.

Violation: Arms swapped roles

- **Visible cue:** the left gripper folds a flap, turns the cradle, lifts a strip, or makes a press stroke; the right gripper braces or pins while the left works; or the left gripper travels to the carton's right end.
- **SOP rule broken:** Steps 1 to 7, the left arm holds and the right arm works, and no strip ever touches the left gripper.
- **Coaching note:** left holds, right works. Both arms move together in Step 7 and nowhere else.

Violation: Reached across instead of turning the cradle

- **Visible cue:** the right gripper works the carton's **left end** (folding the left short flap, laying or pressing the left end strip, or pressing the left overhang) instead of turning the cradle first; or

the arm leans out over the carton to reach the far side of a flap or a seam.

- **SOP rule broken:** Steps 2, 5, and 6, when the job is at the other end the cradle is turned a half turn and the job comes to the arm.
- **Coaching note:** if you are stretching, stop and turn the cradle. The turn is cheaper than the reach and it is what keeps the carton square.

Violation: Half turn not run to the stop

- **Visible cue:** the base is left partly turned, is rocked back and forth, is turned the wrong way along the near edge, or is turned more than once between the same two steps; or the episode ends with the cradle off its home stop.
- **SOP rule broken:** Steps 2, 6, and 8, one continuous swing of the near right tab to the left, to the stop, the same way every time.
- **Coaching note:** one tab, one direction, run it to the stop and look at the stop.

Violation: Cradle turned by the carton

- **Visible cue:** a gripper pushes, twists, or drags the carton itself to turn the cradle, instead of taking the turn tab.
- **SOP rule broken:** Steps 2 and 6, the turn is taken by the tab with both grippers clear of the carton.
- **Coaching note:** the tab turns the cradle. The carton is a passenger.

Violation: Folded or taped without a hold

- **Visible cue:** a flap is folded or a strip is laid and pressed with the left gripper off the carton, or the left gripper takes its pin only after the fold or the lay has started, or comes off before the step is finished.
- **SOP rule broken:** Steps 2 to 6, the pin is taken first and held for the whole step.
- **Coaching note:** pin first, let go last.

Violation: Flaps folded on a tube that is not square

- **Visible cue:** a flap is folded while a corner is still creased flat, a panel is bowed off a channel wall, or an end panel is off its seating line.
- **SOP rule broken:** Steps 1 to 3, the tube is squared before any flap is folded, and a flap is never folded on a tube that is not square.
- **Coaching note:** the tube is squared once, at the start. Everything after it is built on that.

Violation: Fold released before it held

- **Visible cue:** the gripper opens as soon as the flap is down and the flap springs back up, or the episode moves on with a flap standing, curled, or lifting at a corner.
- **SOP rule broken:** Steps 2 and 3, hold the fold flat for 2 seconds before releasing.

- **Coaching note:** hold it down until it stops pushing back.

Violation: Strip taken by the wrong part

- **Visible cue:** a gripper pinches a strip anywhere but its grab tab, touches the sticky face, or peels a strip off the board by dragging it rather than lifting the tab.
- **SOP rule broken:** Steps 4 to 6, the grab tab is the only part of a strip a gripper touches.
- **Coaching note:** tab only. A strip that has touched the gripper is already ruined.

Violation: Strip laid without hanging straight

- **Visible cue:** the strip is curled, twisted, or stuck to itself as it is carried, and it is laid anyway; or a gripper tries to unstick a folded strip in the air instead of scrapping it.
- **SOP rule broken:** Steps 4 to 6, a strip that is not hanging straight is scrapped, not straightened.
- **Coaching note:** look at it hanging before you bring it over. A folded strip is scrap, not a repair job.

Violation: Strip laid off the seam

- **Visible cue:** a strip lands to one side of the seam it should cover, runs at an angle across it, starts short of the end, or leaves the seam line showing along part of its length.
- **SOP rule broken:** Steps 4 to 6, each strip covers its seam end to end.
- **Coaching note:** set the free end on the seam before you travel. The first end decides the whole strip.

Violation: Strip peeled up and stuck down again

- **Visible cue:** a laid strip is lifted off and pressed back down, shifted along the seam after contact, or repositioned by peeling one end.
- **SOP rule broken:** Steps 4 to 6, a badly laid strip is peeled off, scrapped, and replaced with a fresh one.
- **Coaching note:** tape sticks once. Scrap it and take a new strip.

Violation: Press stroke skipped or broken

- **Visible cue:** the episode moves on from a laid strip with no press stroke, lifts the gripper part way along and sets it down again, dabs at the strip in several places, or presses in the wrong direction along an end seam.
- **SOP rule broken:** Steps 4 to 6, one continuous press stroke, end to end, without lifting.
- **Coaching note:** one stroke, all the way across, without coming off the tape.

Violation: Overhang left unpressed

- **Visible cue:** a strip end is left standing out past the carton, hanging loose, or folded back on itself instead of pressed flat onto the panel below it.

- **SOP rule broken:** Steps 4, 5, and 6, every overhang is pressed down onto its panel.
- **Coaching note:** the overhang is what holds the seal. Finish it before you move on.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the peel board, over the bare tabletop, or over open space, and a strip falls, drapes over the carton's side, or lands anywhere but on its seam.
- **SOP rule broken:** Steps 1 to 7, a gripper opens on the carton or on the packing spot and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Strip dragged across the carton

- **Visible cue:** a strip is slid, dragged, or scrubbed along the flaps to position it, or it touches the carton somewhere other than the seam on the way to being laid.
- **SOP rule broken:** Steps 4 to 6, the free end is lowered onto the start of the seam and the strip settles behind the gripper.
- **Coaching note:** lower it onto the seam, do not sweep it there.

Violation: Carton lifted or shifted in the cradle

- **Visible cue:** the carton is lifted, dragged, rotated, or pushed off its seating lines at any point before Step 7, including being pulled out of the channel by a fold or a press stroke.
- **SOP rule broken:** Steps 1 to 6, the carton never leaves the cradle and never comes off its seating lines until Step 7.
- **Coaching note:** the cradle holds the carton square. Anything that moves it undoes Step 1.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides a flap, a strip, or the carton into position after releasing it, instead of taking the prescribed correction.
- **SOP rule broken:** Steps 1 to 7, correct a fold by folding it again, a strip by scrapping it, and a placement by lifting it clear and lowering it again.
- **Coaching note:** do it again properly, or leave it alone.

Violation: Turn over done wrong

- **Visible cue:** the carton is turned with one gripper, rolled away from the front edge, run through both quarter turns in one continuous roll with no set down between them, dropped or tipped the rest of the way instead of lowered, or set down with the taped side up or on its side.
- **SOP rule broken:** Step 7, both grippers hold the end panels and roll the carton toward the front edge one quarter turn, set it down and release, then take it again for the second quarter turn.

- **Coaching note:** both hands, toward you, a quarter at a time, and let it rest in between.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or takes a new hold on a strip or on the carton without setting it down first.
- **SOP rule broken:** Steps 4 to 7, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** a strip falls to the tabletop or floor, the carton is dropped, the cradle or the peel board is struck and shifts, a staged strip is displaced, or the two arms strike each other along the seam line.
- **SOP rule broken:** Steps 1 to 7, keep a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a strip still on the peel board, an overhang unpressed, the carton still in the cradle or taped side up, the cradle off its home stop, an arm away from home.
- **SOP rule broken:** Step 8, confirm the carton, the cradle, and the bare peel board; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Turntable fault:** a stop that does not hold, a base that runs past the stop, or a turn tab that comes off in the gripper.
- **Defective blank:** crushed, torn, damp, creases too stiff to open to a right angle under a straight push, or a tube that springs back out of square once the grippers leave. Replace it before the next episode.
- **Blank cut wrong for the job:** long flaps that do not meet at the center seam, or flaps that overlap so far the seam is not where the SOP says it is.
- **Defective strip:** lost tack, stuck to the peel board, torn on lifting, cut to the wrong length, or a grab tab that has come unfolded. Replace the strip before the next episode.

- **Peel board will not release** a strip in one piece.
- **Cradle or peel board shifts out of its spot** during the episode.

Annotation subtasks (from SOP)

1. Square a flat blank into a tube
2. Fold one flap in
3. Turn the cradle a half turn
4. Lay one tape strip onto a seam
5. Press one tape strip down and press its overhangs
6. Turn the carton a quarter turn and set it down
7. Return both arms home and end the episode